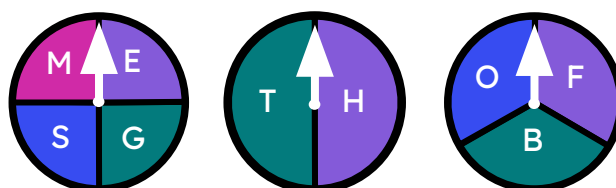




Using lists to display outcomes for more than two events

Task A: Listing outcomes from a multi-stage trial

1) Jun needs to do some revision, so he uses three spinners to decide which **subject** {maths (M), english (E), science (S), geography (G)}, the **duration** {20 mins (T), 60 mins (H)} and which **resource** {Oak National Academy (O), flashcards (F), books (B)}.



a) write down all the possible outcomes for Jun's revision

MTO, ETO, STO, GTO, MTF, ETF, STF, GTF, MTB, ETB, STB, GTB, MHO, EHO, SHO, GHO, MHF, EHF, SHF, GHF, MHB, EHB, SHB, GHB

b) compare your list of possible outcomes to Jun's list of possible outcomes

MTO, EHB, STO, GTO, GHB, ETF, STF, GTF, MTB, ETB, SHB, GTB, MHO, SHO, EHO, MTF, MHF, EHF, STB, GHF, MHB, ETO, STB, GHO

Jun has missed SHF and duplicated STB. His list is not systematic.

2) In a restaurant they have a fixed price three course menu.

Starter
Garlic mushrooms (G)
Halloumi bites (H)
Main
Bacon cheeseburger (C)
Fish and chips (F)
Vegetable curry (V)
Dessert
Ice Cream (I)
Fruit Salad (S)

a) write down all the possible outcomes that include ice cream as the dessert

GCI, GFI, GVI, HCI, HFI, HVI

b) list all the possible outcomes that include vegetable curry as the main course

GVI, HVI, GVS, HVS

c) list all the possible outcomes that include garlic mushrooms as the starter and fruit salad as the dessert

GCS, GFS, GVS



Task B: Listing outcomes from more than two events

1) Andeep, Izzy and Lucas are playing a game using this spinner. Andeep wins a point if the word **starts with a m**, Izzy wins a point if the word has **four letters** and Lucas wins a point if the word **ends in a vowel**.

a) List the outcomes that get Andeep a point.

{mouse, menu, match, milk}

b) List the outcomes that get Izzy a point.

{menu, tree, ship, milk}

c) List the outcomes that get Lucas a point.

{menu, cello, tree, mouse}

d) List the outcomes that get Andeep and Izzy a point.

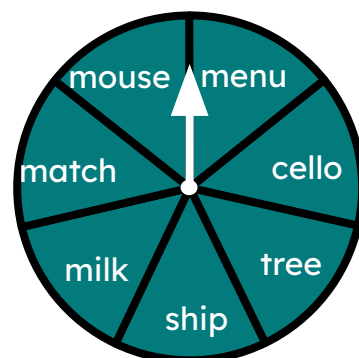
{menu, milk}

e) List the outcomes that get Izzy and Lucas a point.

{menu, tree}

f) List the outcome that gets all of them a point.

{menu}



2) 10 animals are sorted into three categories, whether they are critically endangered, whether they are a mammal and whether they live on land.

Critically endangered = {black rhino, hawksbill turtle, orangutan, vaquita, saola}

Mammal = {black rhino, orangutan, vaquita, blue whale, african wild dog, saola, sea lion, tiger}

Land animal = {black rhino, orangutan, african wild dog, monarch butterfly, macaw, saola, tiger}

a) List all the land mammals that are critically endangered.

{black rhino, orangutan, saola}

b) List all the mammals that are not critically endangered.

{blue whale, african wild dog, sea lion, tiger}

